

Element G

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The pipe we experimented with in school was too thin and cheap. Because of this, we found a pipe at Home Depot that fitted our needs. This pipe was significantly stronger, 2 feet long, and clear so the user can see the water rising.



Unfortunately, the mouthpiece was too small to fit the tube so we decided that we have to design a piece that would fit the PVC pipe and the mouthpiece.



Originally, as a group we wanted to create a barometer that works well with liquid. The image on the right shows the type of pipe that we wanted to use (Design 2 in Element D). However, the device would have to be extremely tall to be able to work and to have the water travel.



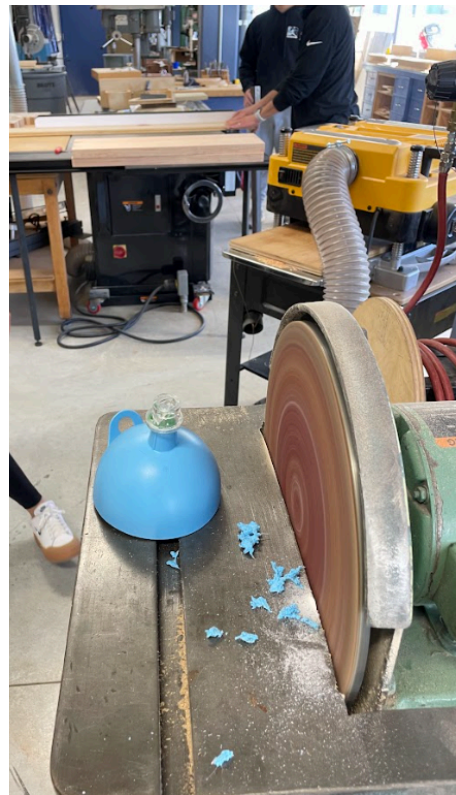
At school, we found this ice cream bucket that held the perfect amount of water and didn't take up too much area. Because of this, we used this bucket in the meantime until we found the actual bucket we wanted to use.



This was our clear pipe before purchasing the official pipe from Home Depot. At first, we just put the pipe in the water and left an opening on the bottom. We then drilled a little hole for the water to go into instead of holding it on an angle. Next, we took a drill and wanted to see what would happen if there was a greater hole in the pipe. We found out it was easier for the water to go up with the bigger hole.

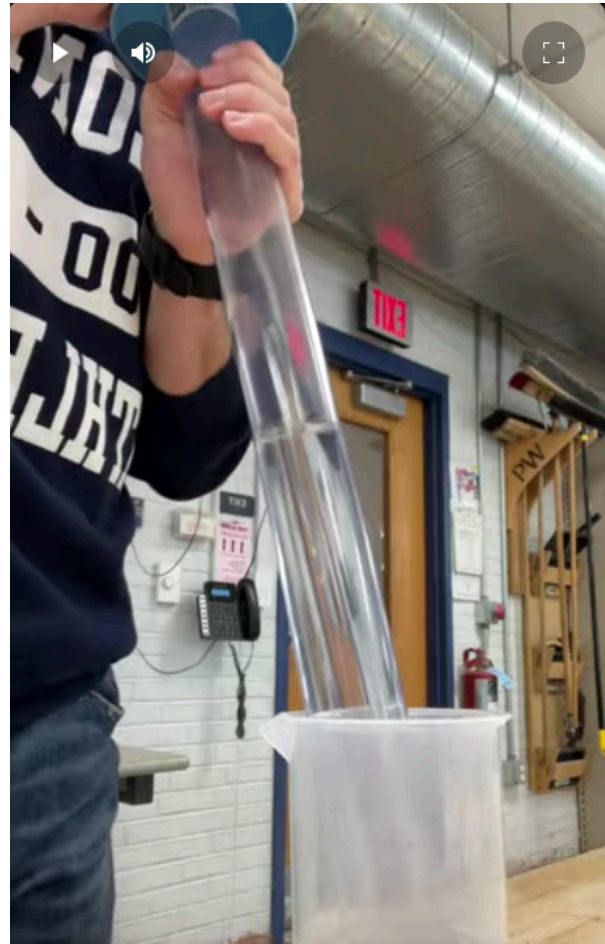


To attach the mouthpiece to the pipe, we decided to use a funnel for the meantime until we designed and printed a piece that would attach the pipe. The top of the funnel was too small for the mouthpiece so we shaved it off to make it fit.

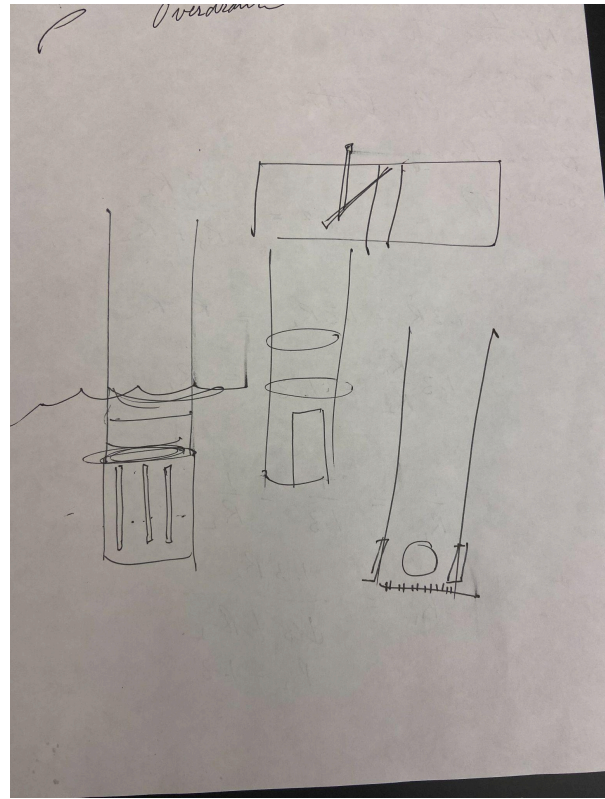


The link and image on the right shows Jack using the initial prototype until we actually made the 3D printed piece

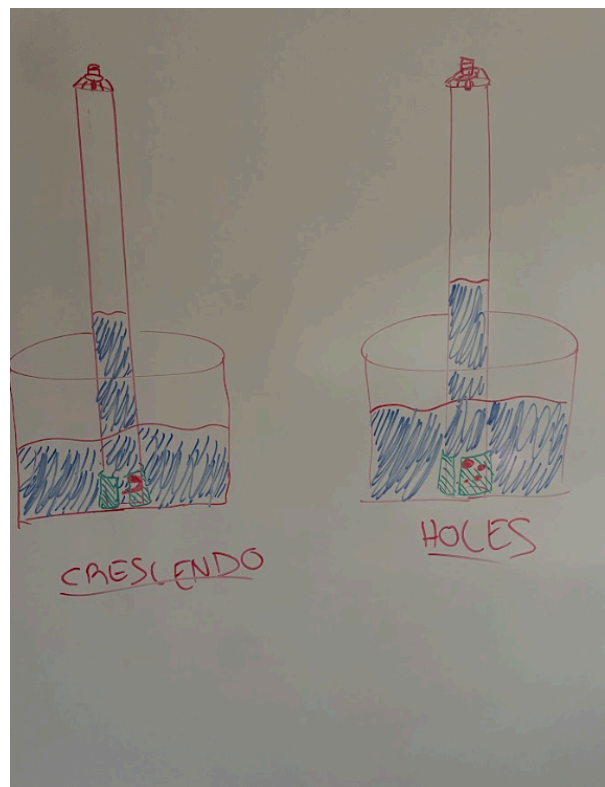
<https://youtube.com/shorts/j5rFUYgRvrg?si=5DfLMbYOxbBtH9FZ>



After testing our initial prototype, we wanted to see what was the most efficient way to get the water smoothly through the pipe. Thus, we went to the physics teacher at our school, Mr. Schidley, and asked him for help. He told us to have some type of gradient behind a slit to restrict how much water would be allowed through.

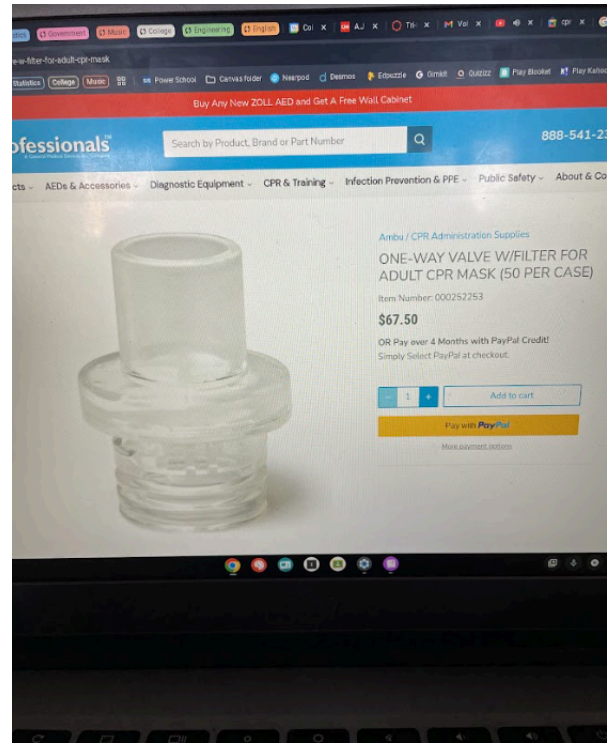


The original design for our PVC pipe was to have different slits at the bottom in which the user can move the 3D printed piece to their own desire. The drawing on the left shows our idea of a crescendo. The pipe would have a big opening at the bottom that would gradually gain more resistance the smaller the opening is, while the 3D printed piece is rotated. The issue with this design is that it is difficult to calibrate. Meaning, it is less accurate as there is no set size. The drawing on the right shows another idea we discussed which was having circles of different sizes, like the

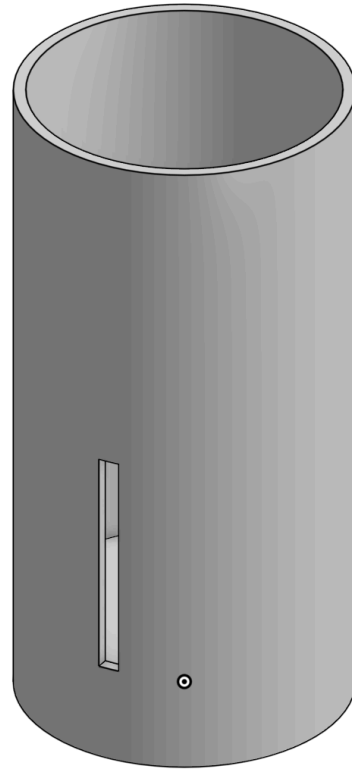


crescendo, the design will use a 3D printed piece attached to the bottom. Similar to the slits, the smaller the circles the higher the resistance is.

The image on the right shows a similar mouthpiece to the one we were testing with. However, this is a pack of 50 and is nearly \$70 which isn't something we can afford so we continued looking.



The image on the right shows the 3D printed piece that Caleb designed. This piece will go on the bottom of the PVC pipe, which contains the big opening. This will help the user control how much water they want rising. Meaning, how much resistance they want; bigger the space the easier it is for more water to rise.



The image on the right shows the 3D printed piece on the pipe. Luckily, it fits really well and at this point we need to make the opening for the crescendo.



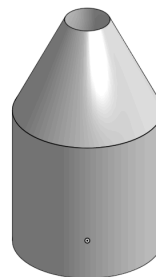
Our original idea was to have a crescendo and this is shown here in Onshape. This crescendo allows water to go through while still being able to control the flow.



While trying to cut the pipe, we figured out that it is hard to get the perfect, smoothest cut. Because of this, our teacher, Mr. Mucrella, took a sharp pair of scissors to smooth it out.



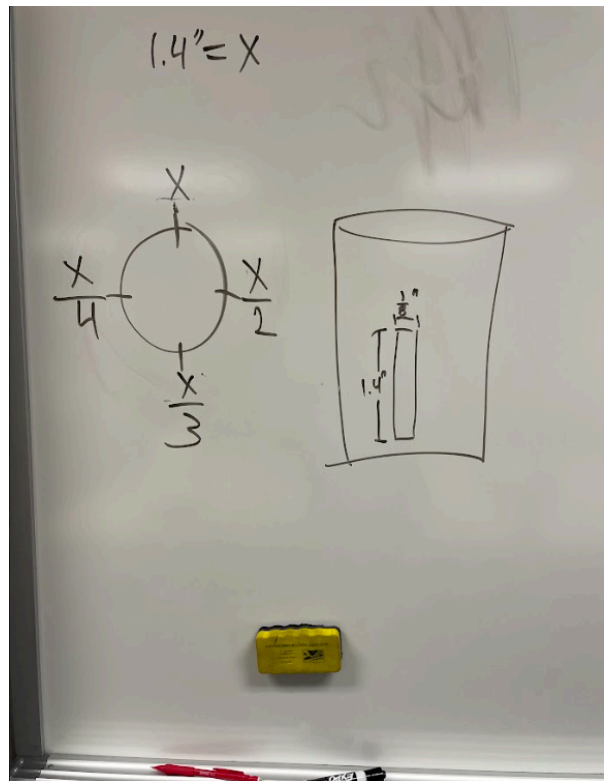
To make a 'funnel' piece for the top of the pipe, we designed a piece in OnShape that would attach to the pipe and fit the mouthpiece.



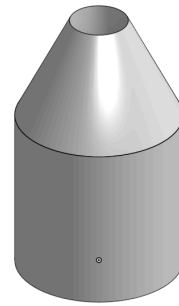
The image on the right shows our final product. The red 3D printed piece is how the mouthpiece is held and where the user will breathe in. Although it might be hard to see, the 3D printed piece containing the slit is at the bottom of the PVC pipe. This is the piece that will be moved to adjust the resistance of the device.



The image on the right shows a concept that Jake and Jack were discussing. The idea was having different slits that were about $\frac{1}{8}$ in apart with varying height so the user could turn the 3D piece to their desired resistance.



We bought new one way valves for the device so multiple users can use it. However, our 3D printed piece was too small for it so we designed a new one but the top is a bit wider to fit the new piece.



This is the final look of our design with the new 3D printed piece containing the new mouthpiece as well.

